# Individual Weekly Report – Week 4

EDA Deep Dive

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| **Department** | Artificial Intelligence and Data Science |
| **Project Title** | Youth Smoking and Drug Consumption: From Raw Data to Visual Insights |
| **Role** | Data Visualization and Dashboard Development |
| **Week** | Week 4 – EDA Deep Dive |

## Objective of the Week

The aim of this week was to explore the dataset in depth through Exploratory Data Analysis (EDA). My part was to create the visualizations that show the patterns and relationships hidden in the data.

## Work Done

**Creating histograms:** I created histograms to understand how the values of different variables are distributed. They helped to see whether the data is evenly spread, concentrated in a small range or skewed to one side.

**Developing boxplots:** I developed boxplots to identify outliers. A boxplot shows the median, the spread of the data and the extreme values in one picture, so it is very useful for finding values that are far from the normal range.

**Creating scatterplots:** I created scatterplots to analyse the relationship between pairs of variables. These plots helped to see whether two variables move together, move in opposite directions or have no clear relationship.

**Studying visual patterns and trends:** After preparing the charts, I studied them carefully to understand the main patterns and trends in the dataset. I noted the important observations so that they could be used in the project discussion and later reports.

## Tools and Techniques Used

Histograms, boxplots and scatterplots, along with careful reading of the patterns they show.

## Outcome of the Week

The charts gave the team a clear visual understanding of the data, including its distributions, outliers and relationships between variables.

## What I Learned

I learned how different plots answer different questions. Histograms show spread, boxplots show outliers and scatterplots show relationships. Using them together gives a much fuller picture of the data.

## Contribution to the Project

My EDA visualizations became the main visual evidence for the team's analysis and were useful for the upcoming mid-review.

## Challenges Faced

With many variables in the dataset, it was difficult to decide which plots would be most useful. I handled this by choosing the plot type according to the question I wanted to answer, such as spread, outliers or relationship.

## Plan for Next Week

In Week 5, I will prepare the mid-review slides, organize the graphs created so far and present the initial results.

## Weekly Summary

Week 4 gave the project a strong visual understanding of the data. The histograms, boxplots and scatterplots became the main evidence of our analysis.